Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/12/2006
PI Name:	Purdy, Ralph Ph.D.		
Project Title:	Effect of Microgravity on Vascular Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	repurdy@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-7653
Organization Name:	University of California - Irvine		
PI Address 1:	Department of Pharmacology, College of Medicine		
PI Address 2:	350C Med Surge II, zot 4625		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-4625	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE through 9/30/2007, per A. Chu (8/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK29G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Hindlimb unweighting in the rat was used to simulate microgravity and arteries were isolated to study the effects of HU on vessel function, morphology and compliance. Evidence was obtained that HU: 1) decreases the expression of myosin heavy chain (MHC), myosin light chain 20 (MLC-20) and alpha actin protein expression and that this could account for the HU-induced vasoconstrictor hyporesponsiveness of rat abdominal aorta; 2) decreases MLC-20 protein expression and vessel wall compliance, but does not change morphometric parameters in the carotid artery; 3) decreases both alpha actin and MLC-20 protein expression, increases vessel wall compliance, but does not change morphometric parameters in the femoral artery; 4) causes a marked up-regulation of the contribution of nitric oxide mechanisms to the vasoconstrictor response of thoracic aorta; 5) has no effect on alpha ARs or alpha 2ARs in any vessel studied.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The following are the accomplishments of the reporting period. These have been published as abstracts (see next section; BIBS) and have been submitted as full articles for publication in peer-reviewed journals or are to be submitted as full articles during April, 2006. Thus, authors, manuscript titles and journals are indicated at the end of each project description. HU decreases the expression of myosin heavy chain (MHC), myosin light chain 20 (MLC-20) and ?actin protein expression in abdominal aorta. The goal of this study was to investigate the effect of microgravity on the expression of the myofilament proteins alpha actin, myosin light chain 20 (MLC-20), myosin heavy chain (MHC), and the myofilament binding proteins heavy caldesmon, calponin, and alpha tropomyosin in rat abdominal aortic smooth muscle. Microgravity was simulated by hindlimb unweighting (HU) Wistar rats for 20 days. Western blots revealed an HU-induced downregulation of alpha actin, MLC-20, and MHC, an upregulation of heavy caldesmon, and no change in calponin and alpha tropomyosin. Quantitative Real Time RT-PCR measurements showed no difference in mRNA expression for alpha actin or MHC, but a decrease in that for MLC-20. Total arterial DNA was also not different between Control and HU. We hypothesize that the HU-induced decrease in contractile proteins plus the increase in the actin binding protein, heavy caldesmon, could contribute to the vascular hyporesponsiveness observed in HU-treated abdominal aorta. Heavy caldesmon, calponin and alpha tropomyosin characteristically decrease during vascular smooth muscle cell dedifferentiation. Because the levels of these proteins were either unchanged or increased (heavy caldesmon), the HU-induced reductions of alpha actin, MHC and MLC-20 were not caused by classical smooth muscle dedifferentiation. Moreover, the lack of change in total DNA argues against an HU-induced hypoplasia. The reduction in MLC-20 was due, in part, to HU-reduced transcriptional activity while the HU-induced reductions of alpha actin and MHC appear to have occurred at the translational level. This work is published as an abstract and has been submitted for publication in the Journal of Applied Physiology: Hwang,S., Sun, A., Haddad, F., Goraya, N., Huynh, J., Sang, S., Trang Nguyen, T Willis, V., Purdy, R.E. Simulated Microgravity Alters Vascular Alpha Actin, Myosin, and Heavy Caldesmon. J Applied Physiol Submitted, 2006. Contractile protein expression, compliance and morphometric parameters in the carotid and femoral arteries. The goal of this study was to determine the effects of microgravity on myofilament protein expression and both passive and active length-force relationships in carotid and femoral arteries. Microgravity was simulated by 20-day hindlimb unweighting (HU) in Wistar male rats, and carotid and femoral artery segments were isolated from both HU and control (CTL) rats for western blot and length-force analysis. Western blots revealed that HU significantly decreased myosin light chain 20 (MLC-20) protein levels in both carotid and femoral arteries, and decreased myosin heavy chain (MHC) in femoral artery. Alpha actin levels were not altered by HU treatment in either artery bed. Length-force analysis demonstrated that HU did not change either passive or active length-force relationships in the femoral artery. HU-treated arterial rings did develop significantly less force to 100 mM K+ than CTL, however optimal lengths were identical. In the carotid artery, length active-force curves were identical for both CTL and HU; however the length passive-force curve for HU-treated rings exhibited a steeper slope than CTL, suggesting decreased compliance of the artery wall. In conclusion, these data suggest that the HU-induced decreases in both MLC-20 and MHC in femoral artery are responsible for the decreased contraction to 100mM K+ in HU-treated femoral artery rings. In the carotid artery, the HU-induced decrease in vessel wall compliance may counter any decrease in contractility caused by the decreased MLC-20 levels. This work is in the final stages of preparation for submission to the Journal of Applied Physiology: Hwang,S., Shelkovnikov, S.A., and Purdy, R.E. The Effects of Simulated Microgravity on the Rat Carotid and Femoral Arteries: The Role of Contractile Protein Expression and Mechanical Properties of the Vessel Wall. To be submitted, J Applied Physiol, 2006. HU causes a marked up-regulation of the contribution of nitric oxide mechanisms to the vasoconstrictor response of thoracic aorta. In this study, it was found that the decreased vasoconstrictor responsiveness of thoracic aorta was reversed by mechanical removal of the endothelium and led to an even greater contraction than in control tissues. This difference may be due to an increased sensitivity of the endothelium to muscarinic agonists such as methylfurmethide and there may also be an increased sensitivity of the vascular smooth muscle to nitric oxide. Furthermore there was a greater contractility deficit in Sprague-Dawley rats exposed to simulated microgravity as compared to Wistar rats. This work is in the final stages of preparation for submission to the Journal of Applied Physiology: Summers, S.M., Ulloa, A.S., and Purdy, R.E. Simulated Microgravity Induces Nitric Oxide Related Changes in the Rat Thoracic Aorta. To be submitted, J Applied Physiol, 2006. HU has no effect on alpha 1ARs or alpha 2ARs in any vessel studied. Simulated microgravity causes a marked depression of the maximal vasoconstrictor response of rat abdominal aorta to norepinephrine (NE). The goal of the this study was to determine if this depression of vasoconstriction was mediated by an effect on alpha adrenoceptors (alplha ARs). Microgravity was simulated by hindlimb unweighting (HU) for 20 days. Alpha ARs were studied by measuring isometric contractions of vessel rings before and after partial inactivation of the Alpha AR population with phenoxybenzamine (POB), or in the presence and absence of selective alpha AR antagonists. Concentration response curves (CRCs) before and after POB treatment were analyzed to determine affinity (KA) and efficacy (percent receptor occupancy at EC50). Both NE log EC50 values, 7.65+0.1 and 7.7+0.1, and log KA values, 6.15+0.2 and 6.14+0.2 in control and HU aorta rings, respectively, were not significantly different. Respective receptor occupancies at EC50, 1.83 and 1.7% also were not different. Similar results were obtained using phenylephrine as the agonist. Real Time RT-PCR revealed an HU-induced decrease in the mRNA for alpha1B-ARs, but no change in that for alpha1A-ARs or alpha1D-ARs. NE CRCs were obtained in the presence and absence of either the alpha1D-AR selective antagonist, BMY7378 or the alpha2-AR antagonist, rauwolscine. The magnitudes of blockade by these antagonists were equal in control versus HU aorta. In conclusion, NE elicits a contraction in rat abdominal aorta acting on both alpha1D-ARs and alpha2-ARs. Simulated microgravity has no effect on either of these receptor subtypes. This work has been submitted for publication in the Journal of Pharmacology and Experimental Therapeutics: Shelkovnikov, S.A., Sun, A., Summers, S.M., and Purdy, R..E. Effect of Simulated Microgravity on Vascular Alpha Adrenoceptors. J Pharmacol Exper Ther Submitted, 2006.
Bibliography Type:	Description: (Last Updated: 12/28/2007)

Abstracts for Journals and Proceedings	Hwang, S., Goraya, N., Willis, V., Chan, D. Miao, A., Ali, A., Purdy, R.E. "Simulated microgravity alters vascular alpha-actin, myosin and heavy caldesmon." American Society of Gravitational and Space Biology annual mtg, Reno, NV, November 2005. Gravit Space Biol 2005 Nov;19(1):10. , Nov-2005
Abstracts for Journals and Proceedings	Summers, S.M., Ulloa, A.S., Purdy, R.E. "Simulated microgravity in Wistar rats markedly increases vasoconstrictor capacity in endothelium denuded thoracic aorta." American Society of Gravitational and Space Biology annual meeting, Reno, NV, November 2005. Gravit Space Biol 2005 Nov;19(1):10. , Nov-2005
Abstracts for Journals and Proceedings	Shelkovnikov, S., Sun, A., and Purdy, R.E. "Effect of simulated microgravity on vascular a adrenoceptor function." Experimental Biology meeting, April 2006. FASEB J 2006 Apr;20:A1252 , Apr-2006
Abstracts for Journals and Proceedings	Hwang, S., Shelkovnikov, S., and Purdy, R.E. "Effect of simulated microgravity on contractility, contractile protein expression and vascular compliance." Experimental Biology meeting, April 2006. FASEB J 2006 Apr;20:1253A. , Apr-2006